Level 2
Hackathon – Fundamentals of Web Development

Student Details

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Project Title: **Online Resume Builder**

Git-Hub Link:

https://github.com/<your-username>/online-resume-builder *(replace with your repository link)*

1. Problem Statement

Students and fresher candidates often need a resume urgently for internships, campus placements and job applications, but they do not have a clear format and do not know how to organise their information. Word-processor templates are heavy, need software installation, and often break formatting; many online resume sites are paid or demand sign-up.

There is therefore a need for a simple, free, lightweight web application that runs in any browser, lets the user enter their personal, educational, skill, project and certification details in one clean form, and lets them save the information as a PDF without installing anything. The Online Resume Builder addresses this need.

2. Objectives

- To build a responsive, easy-to-use web form that collects all the details required for a resume (name, career title, contact, education, skills, projects, certifications, career objective).
- To apply the fundamentals of web development: semantic HTML for structure, CSS for styling and responsiveness, and JavaScript for behaviour.
- To let the user save or print the entered information as a PDF with a single click using the browser's print feature.
- To provide a one-click "Clear Form" option so a new resume can be started quickly.
- To ensure the application works on both mobile phones and desktops without any backend, database or installation.

Expected outcome: a working single-page application that helps a student prepare resume information quickly and export it as a PDF.

3. Architecture

The application is a client-side, single-page web application contained in one file (`resume_builder_without_preview.html`). There is no server or database; everything runs in the user's browser.

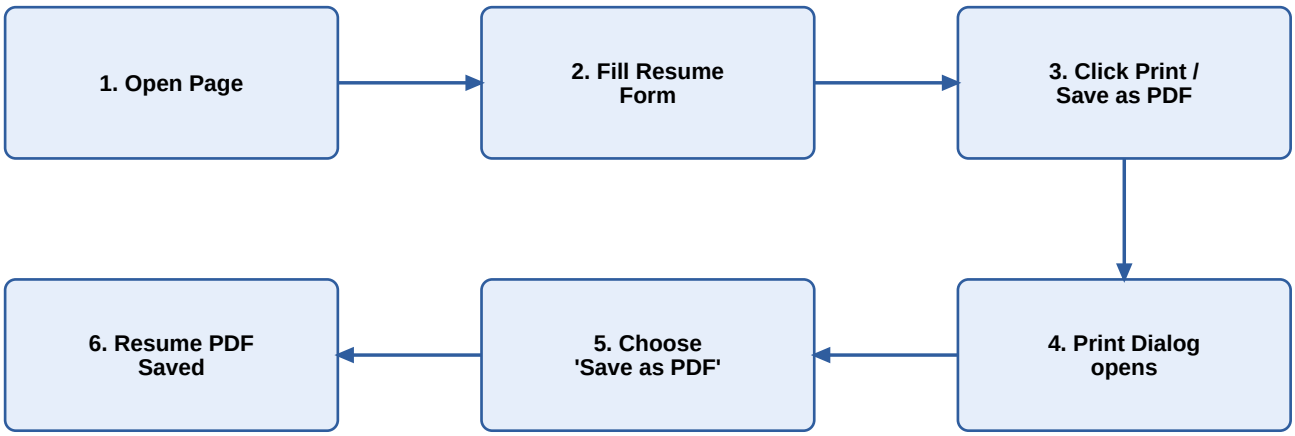
Application structure

- **Header** – title "Online Resume Builder" with a short instruction line.
- **Main / Card section** – the "Student Details" card which holds the form.
- **Form (#resumeForm)** – 8 input fields / text areas, a Print / Save as PDF button and a Clear Form (reset) button.
- **Note** – a helper message explaining how to save the PDF.

Technologies and their roles

Technology	Role in the application
HTML5	Creates the page structure: header, main, section, form, labels, inputs, textareas and buttons.
CSS3	Styles the page: colours, card layout, spacing, form controls, buttons, and the media query that makes it mobile-friendly.
JavaScript	Adds behaviour: attaches a click event to the Print button and calls window.print() to open the PDF/print dialog.
Browser Print Engine	Converts the page with the entered data into a PDF when the user chooses "Save as PDF".

User navigation / application flow diagram



The user opens the page, fills the form fields, clicks **Print / Save as PDF**, the browser print dialog opens through `window.print()`, and the user chooses "Save as PDF" to obtain the final output. The **Clear Form** button can be used at any time to reset all fields.

4. Practical Implementation

The application was developed step by step:

- **Step 1 – Structure:** created the HTML5 skeleton with a header, a main container and a card section; added a form with labelled inputs for every resume detail.
- **Step 2 – Styling:** wrote CSS to give a blue header, a white rounded card on a light background, full-width inputs with borders, and bold blue buttons. A reset-style grey button was added for "Clear Form".
- **Step 3 – Responsiveness:** used `box-sizing:border-box`, a max-width container and a media query for screens up to 500px so the layout fits phones.

- **Step 4 – Logic:** added one JavaScript event listener on the Print button which calls `window.print()` so the entered data can be saved as a PDF.
- **Step 5 – Testing:** tested in a desktop browser and on a mobile browser with sample data; verified Save as PDF and Clear Form.

4.1 Features Implemented

Feature	Purpose
Personal details fields	Collect full name, career title, phone number, email and city/address for the resume header.
Education field	Records degree, college, year and marks.
Skills field	Lists technical and soft skills (HTML, CSS, communication ...).
Projects field	Describes academic / personal projects with short descriptions.
Certifications field	Stores certificate names and issuing organisations.
About Me / Career Objective	Gives a short introduction or objective statement.
Required-field validation	The Full Name field is marked required; phone and email inputs use proper types (tel, email) for correct keyboards and format checking.
Print / Save as PDF button	Opens the print dialog with one click so the user can save the data as a PDF.
Clear Form button	Resets every field instantly to start a new resume.
Responsive design	Layout adapts to mobile and desktop screens.
No installation / no backend	Runs fully offline in any modern browser; user data never leaves the device.

4.2 Source Code

The complete source code is in a single file. The important sections and their functionality are shown below.

HTML

HTML structures the content of the application. The `<header>` holds the title, `<main>` and `<section class="card">` hold the content, and the `<form>` contains `<label>`, `<input>` and `<textarea>` elements for each resume detail, plus two buttons.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Online Resume Builder</title>
<style> ...see CSS section... </style>
</head>
<body>
<header><h1>Online Resume Builder</h1><p>Enter your details to prepare your resume
    information</p></header>
<main>
<section class="card">
<h2>Student Details</h2>
<form id="resumeForm">
<label for="name">Full Name</label><input id="name" required placeholder="Enter your
    full name">
<label for="role">Career Title</label><input id="role" placeholder="e.g. College
    Student">
<label for="phone">Phone Number</label><input id="phone" type="tel" placeholder="Enter
    phone number">
<label for="email">Email Address</label><input id="email" type="email"
    placeholder="Enter email address">
<label for="address">Address / City</label><input id="address" placeholder="Enter your
    city">
<label for="education">Education</label><textarea id="education" placeholder="Degree,
    college, year, marks"></textarea>
<label for="skills">Skills</label><textarea id="skills" placeholder="HTML, CSS,
    communication..."></textarea>
<label for="projects">Projects</label><textarea id="projects" placeholder="Project name
    and short description"></textarea>
<label for="certs">Certifications</label><textarea id="certs" placeholder="Certificate
    name and organization"></textarea>
<label for="summary">About Me / Career Objective</label><textarea id="summary"
    placeholder="Write a short introduction"></textarea>
<button type="button" id="printBtn">Print / Save as PDF</button>
<button type="reset" class="secondary">Clear Form</button>
<p class="note">This simple version does not show a resume preview. Use Print / Save as
    PDF to print this page or save the entered information as a PDF.</p>
</form>
</section>
</main>
<script> ...see JavaScript section... </script>
</body>
</html>
```

CSS

CSS styles the application: it resets box sizing, sets the page font and background, designs the blue header, the white card with rounded corners and shadow, form controls, buttons, and a media query for small screens.

```
* {
    box-sizing:border-box
}

body {
    margin:0;
    font-family:Arial,sans-serif;
    background:#f2f5fb;
    color:#222
}
```

```
header {
    background:#2457d6;
    color:white;
    padding:24px 16px;
    text-align:center
}

header h1 {
    margin:0 0 7px;
    font-size:27px
}

header p {
    margin:0
}

main {
    max-width:680px;
    margin:24px auto;
    padding:0 14px
}

.card {
    background:white;
    padding:22px;
    border-radius:12px;
    box-shadow:0 3px 14px #0000000d
}

h2 {
    margin:0 0 14px;
    font-size:20px;
    color:#2457d6
}

label {
    display:block;
    font-weight:bold;
    font-size:14px;
    margin:13px 0 5px
}

input,textarea {
    width:100%;
    padding:11px;
    border:1px solid #ccd3df;
    border-radius:7px;
    font:inherit
}

textarea {
    min-height:78px;
    resize:vertical
}

button {
    border:0;
    border-radius:7px;
    padding:12px 15px;
    background:#2457d6;
    color:white;
    font-weight:bold;
    cursor:pointer;
    margin-top:16px;
    width:100%;
    font-size:15px
}

.secondary {
    background:#667085
}

.note {
    font-size:13px;
    color:#667085;
```

```
    line-height:1.5
}

@media (max-width:500px) {
    .card {
        padding:16px
    }

    header h1 {
        font-size:23px
    }
}
```

JavaScript

JavaScript implements the application logic. It finds the Print button by its id and attaches a click listener (using an arrow function) that calls `window.print()`, which opens the browser's print dialog where the user can choose "Save as PDF".

```
document.getElementById('printBtn').addEventListener('click',()=>window.print());
```

4.3 Project Execution

- Save the code as `resume_builder_without_preview.html` (or clone the GitHub repository).
- Double-click the file (or right-click → Open with → Chrome / Edge / Firefox). No server or installation is needed. In VS Code, the "Live Server" extension can also be used.
- Fill in the form: name, career title, phone, email, city, education, skills, projects, certifications and career objective.
- Click **Print / Save as PDF**. The browser print dialog opens.
- Select **Save as PDF** as the destination and click Save. The entered information is saved as a PDF file.
- Click **Clear Form** to erase all fields and enter new details.

Working process (input to output): user input in form fields → click on Print button → JavaScript click event → `window.print()` → browser generates PDF → PDF output.

5. Output Screenshots

The output of the application is the Online Resume Builder page showing a blue header with the title, followed by the "Student Details" card containing the form fields (Full Name, Career Title, Phone Number, Email Address, Address/City, Education, Skills, Projects, Certifications, About Me / Career Objective) and the "Print / Save as PDF" and "Clear Form" buttons. After the user clicks Print, the browser print dialog appears and the filled information is saved as a PDF.

6. Technical Explanation

- **Semantic HTML & forms:** `label for=id` links each label to its input for accessibility; `required`, `type="email"` and `type="tel"` give built-in browser validation and suitable mobile keyboards; `type="reset"` clears the form without extra code.
- **CSS box model & layout:** `box-sizing:border-box` keeps widths predictable; `max-width` with `margin:auto` centres the card; border-radius and box-shadow give the card look.
- **Responsive design:** the `@media(max-width:500px)` query reduces padding and heading size on phones; the viewport meta tag enables correct scaling.
- **DOM & events:** `document.getElementById()` selects the button element from the DOM and `addEventListener('click', ...)` handles the user's click using an arrow function.

- **Browser Print API:** `window.print()` invokes the browser's native print / save-as-PDF feature, so no external PDF library is required.
- **Client-side only:** there is no server code, so data stays on the user's device, giving privacy and offline use.

7. Conclusion

The Online Resume Builder successfully provides a simple, fast and free way for students to enter their resume details and save them as a PDF. Through this project the fundamentals of web development were practised: structuring a page with HTML forms, styling and making it responsive with CSS, and adding interaction with JavaScript events and the browser Print API. The application works on any device with a browser and demonstrates how a useful tool can be built with only front-end technologies.

8. Future Enhancements

- Add a **live resume preview** and multiple professional templates (colour and layout choices).
- Add profile photo upload and sections such as experience, achievements, languages and social/LinkedIn links.
- Save form data using **localStorage** so information is not lost on refresh, and allow export/import as JSON.
- Direct **PDF / DOCX download** using libraries such as jsPDF, without using the print dialog.
- User login with a secure backend and database for saving multiple resumes, with encrypted storage for better security.
- AI-based suggestions for career objective and skill descriptions, ATS-friendly checks, and multi-language support.
- Improved user experience: dark mode, drag-and-drop section ordering, form validation messages and accessibility improvements.